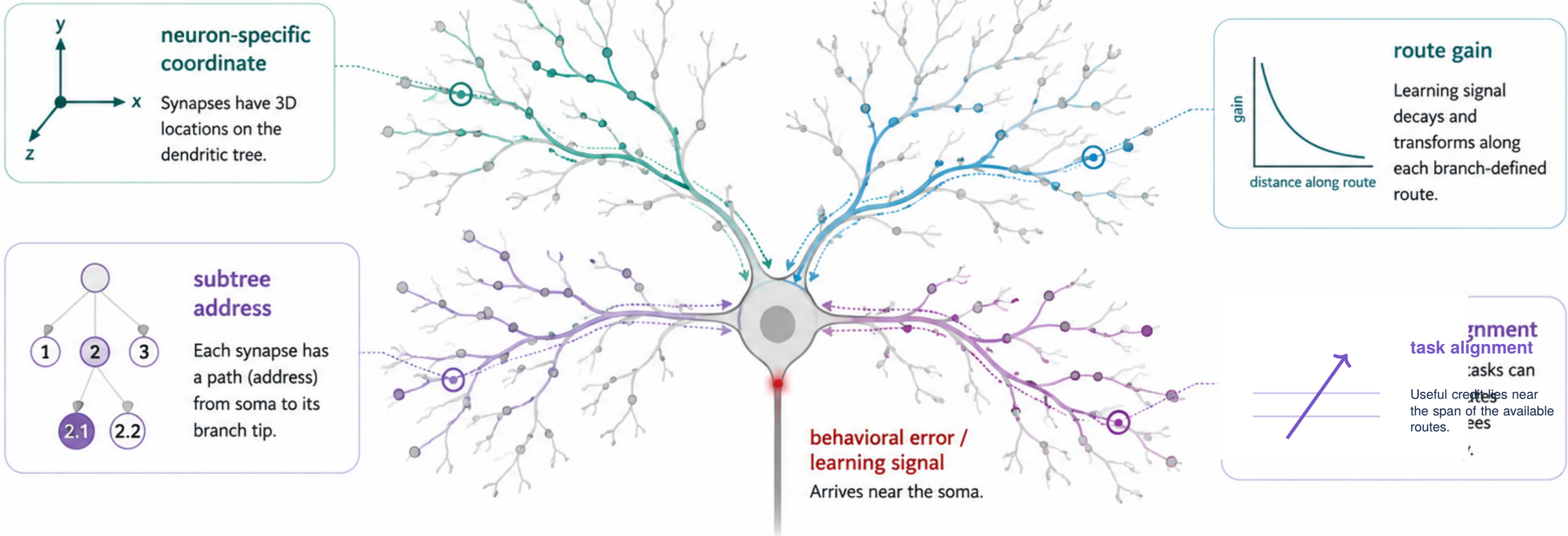


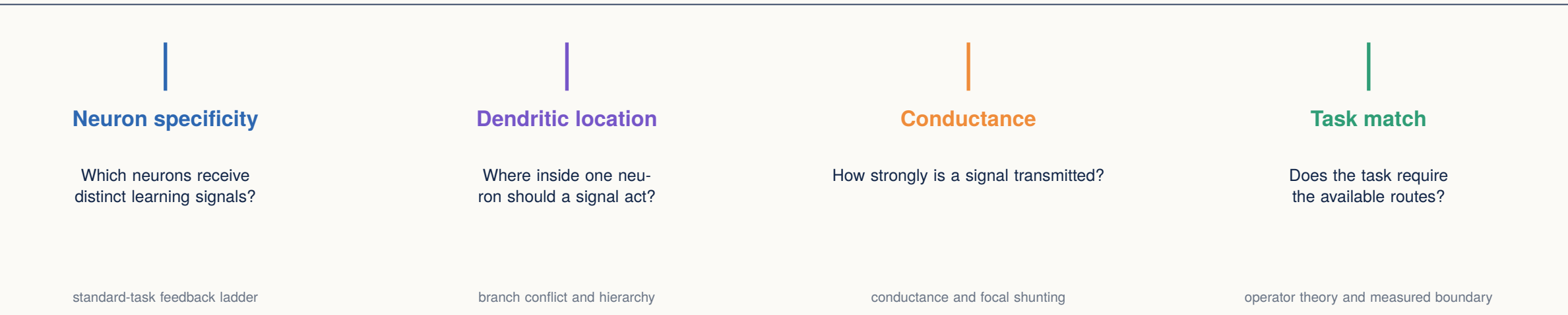
When dendritic structure helps local credit assignment

Coordinates, subtree addresses, route gain, and task alignment



A conditional theory of coordinate, address, gain, and task-route alignment

Four separable questions organize the evidence



EVIDENCE PATH



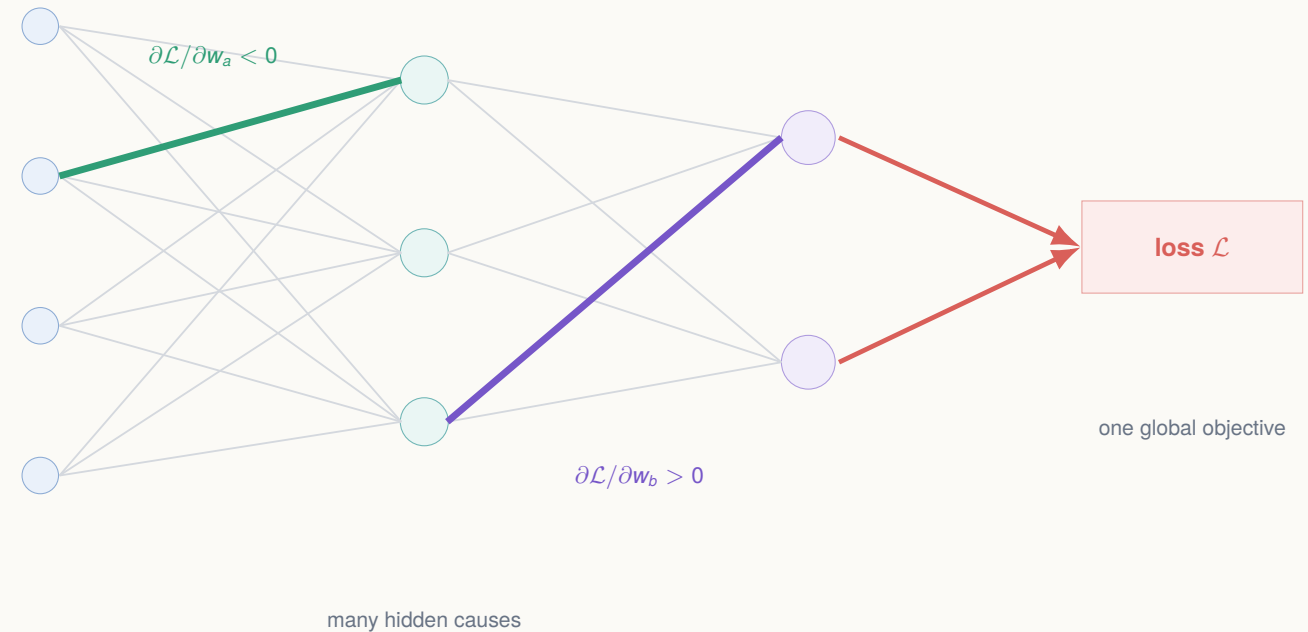
The experiments test progressively stronger claims: route availability, mechanistic sufficiency, and endogenous use are not the same.

A network objective does not identify its local causes

A single loss evaluates the network output. Learning must assign that loss to many internal parameters, whose useful changes can differ in sign and magnitude.

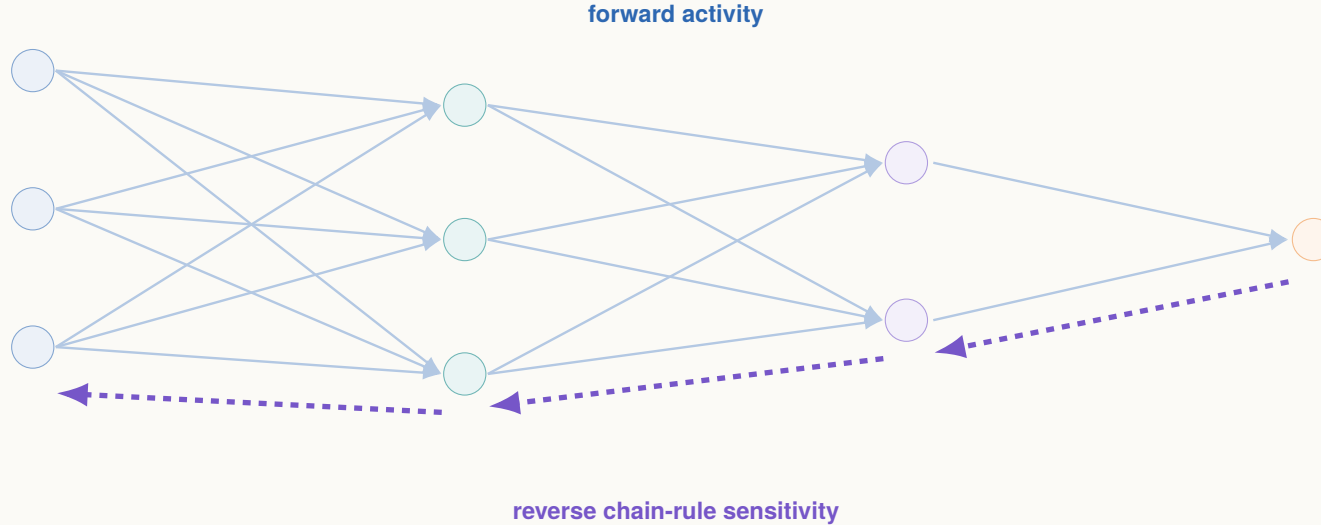
$$\mathbf{w}^+ = \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathcal{L}$$

Credit assignment asks, for each weight: which destination, which sign, and what magnitude?



Learning is not only changing weights; it is assigning a task-dependent direction to each internal weight.

Backpropagation provides the exact computational reference



$$\frac{\partial \mathcal{L}}{\partial w_{ij}} = \underbrace{x_j f'_i(u_i)}_{\text{local eligibility } e_{ij}} \underbrace{\delta_i}_{\partial \mathcal{L} / \partial a_i}$$

$$\delta_i = \sum_{k:i \rightarrow k} w_{ik} f'_k(u_k) \delta_k$$

The eligibility is locally tied to a parameter. The difficult quantity is δ_i : the task-dependent effect of node i on the downstream loss.

Exact destination

each hidden node receives its own derivative

Exact direction

the sign follows the downstream chain rule

Reference, not mechanism

BP defines the information ceiling used below

From BP to local to dendritic-local learning

exact backpropagation

$$\nabla_{w_{ij}} \mathcal{L} = e_{ij} \delta_i$$

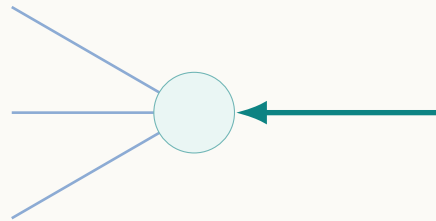


exact reverse recursion

Eligibility is local; the returned error is exact and layer coordinated.

point-neuron local rule

$$\Delta w_{ij} = -\eta e_{ij} \hat{\delta}_i$$

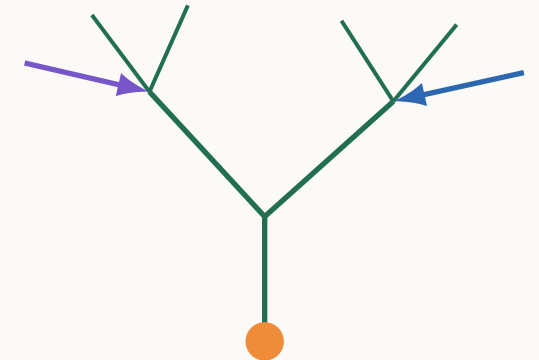


one approximate coordinate per neuron

Eligibility remains parameter specific; feedback is restricted at neuron resolution.

dendritic-local rule

$$\Delta g_{i,n} = -\eta e_{i,n}^{\text{den}} \hat{\delta}_{n,u}^V$$



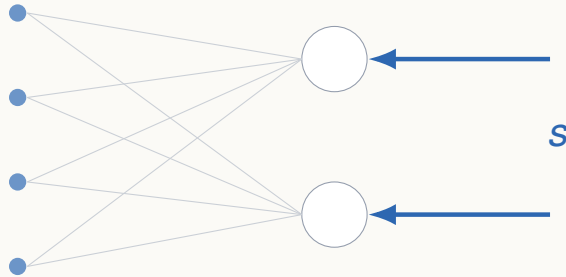
several spatially addressed coordinates

The same factorization is extended to conductances and locations inside one arbor.

Sources of $\hat{\delta}$ in prior work include three-factor rules, feedback alignment, equilibrium or predictive methods, and dendritic-error models.

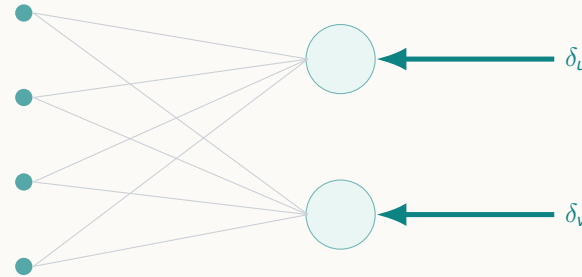
Neuron identity is the first feedback bottleneck

layer-wide scalar



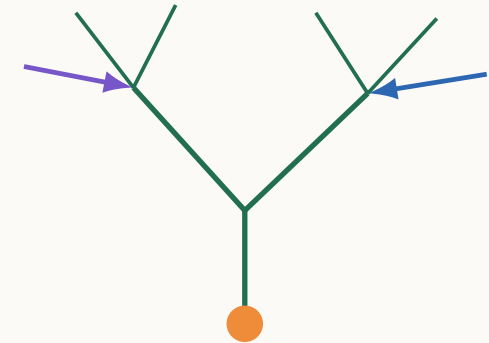
s repeats across the layer; each eligibility e_{ui} remains different

one coordinate per neuron



the signal now identifies the correct neuron

within-neuron address



opposing branch updates require more than neuron identity

$$\Delta w_{ui} = -\eta e_{ui} \hat{\delta}_u$$

A shared scalar does not collapse neurons: distinct inputs, states, and eligibilities preserve distinct weights. It compresses the task-dependent part of learning into one direction.

The standard bandwidth question is which neuron; the dendritic question is where within that neuron.

A dendritic tree adds local state, subtree addresses, and route gain

local state

Voltage and conductance make eligibility compartment and state dependent.

subtree address

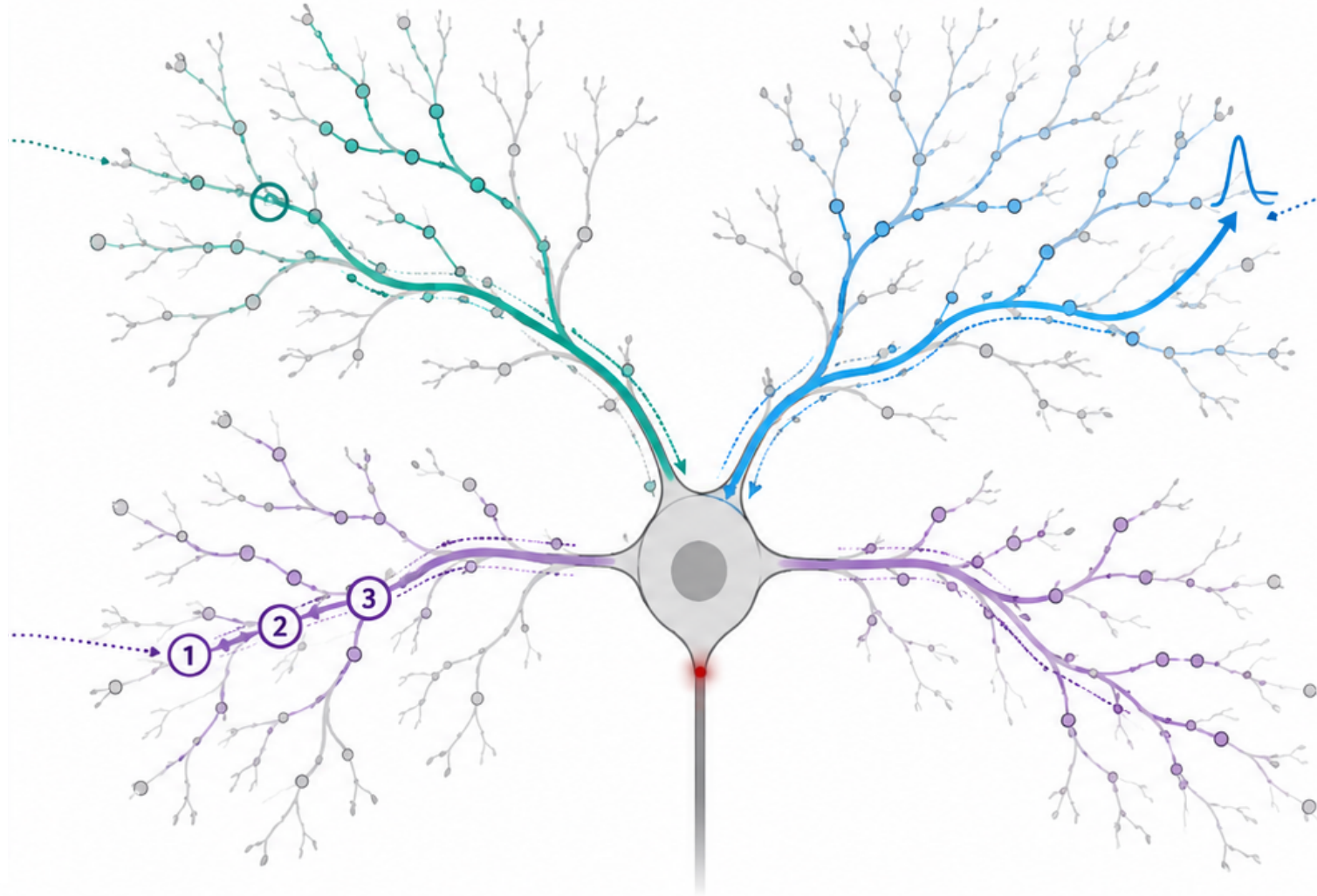
One routed signal can reach a branch and its descendants without being broadcast to the full arbor.

route gain

Attenuation and local conductance regulate how strongly a signal is transported along a route.

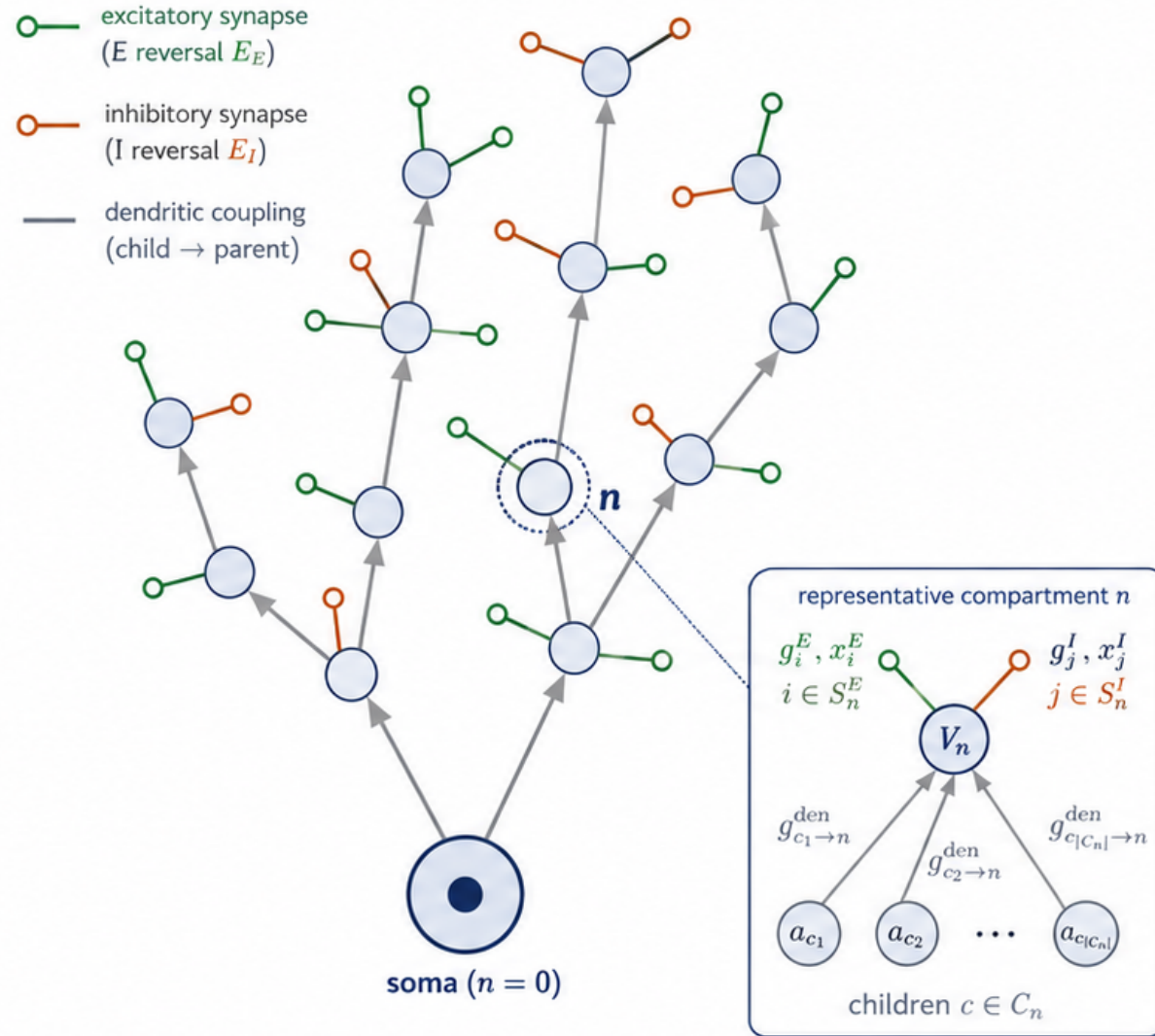
task alignment

These resources help only when the required credit field lies near their available span.



Dendrites make spatial state and candidate credit routes explicit; they do not guarantee a learning benefit.

Excitation and inhibition differ through reversal potential, while both load the membrane



1) Synapse sets and aggregates at compartment n

$$S_n^E, S_n^I, \quad S_n = S_n^E \uplus S_n^I$$

$$G_n^E = \sum_{i \in S_n^E} g_i^E x_i^E$$

$$G_n^I = \sum_{j \in S_n^I} g_j^I x_j^I$$

$$G_n^{\text{child}} = \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}}$$

$$J_n^{\text{child}} = \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}} a_c$$

2) Current balance and steady-state solution

$$C_n \frac{dV_n}{dt} = g_n^L (E_L - V_n) + G_n^E (E_E - V_n) + G_n^I (E_I - V_n) + \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}} (a_c - V_n)$$

$$g_n^{\text{tot}} = g_n^L + G_n^E + G_n^I + G_n^{\text{child}} \quad R_n^{\text{tot}} = \frac{1}{g_n^{\text{tot}}}$$

$$V_n = R_n^{\text{tot}} (g_n^L E_L + G_n^E E_E + G_n^I E_I + J_n^{\text{child}})$$

3) Normalized leaf approximation

$$g_n^L = 1, \quad E_L = E_I = 0, \quad E_E = 1$$

$$V_{\text{leaf}} = \frac{G^E}{1 + G^E + G^I}$$

4) Interpretation

Excitatory and inhibitory synapses both add positive conductance to the denominator; their different reversal potentials determine the sign of current.

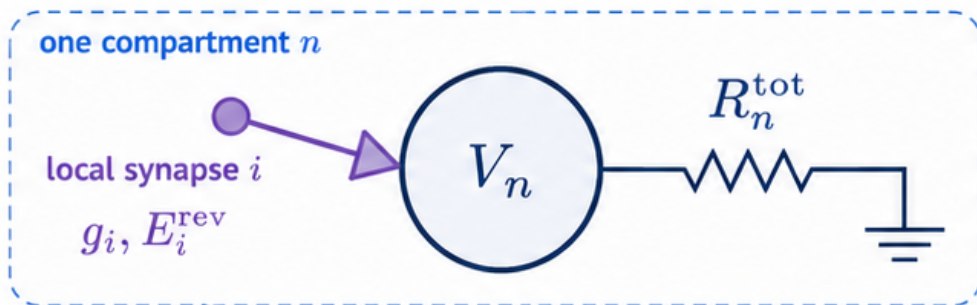
Exact local eligibility with explicit E/I synapses

Exact gradient factorization at synapse i on compartment n

$$\frac{\partial L}{\partial g_i} = e_i^{\text{den}} \delta_{n,u}^V$$

$$e_i^{\text{den}} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)$$

$$\delta_{n,u}^V = \frac{\partial L}{\partial V_n}$$



Local eligibility factors e_i^{den}

presynaptic activity

$$x_i^X$$

transmitter release
or spike-dependent
term

local input resistance

$$R_n^{\text{tot}}$$

total input resistance
seen at compartment n

driving force

$$(E_X - V_n)$$

difference between
reversal potential of
class X and local voltage

Sensitivity of V_n to a synaptic conductance of class X

$$\frac{\partial V_n}{\partial g_i^X} = x_i^X R_n^{\text{tot}} (E_X - V_n), \quad X \in \{E, I\}$$

Explicit excitatory and inhibitory gradient cases

$$\frac{\partial L}{\partial g_i^E} = x_i^E R_n^{\text{tot}} (E_E - V_n) \delta_{n,u}^V, \quad i \in \mathcal{S}_n^E$$

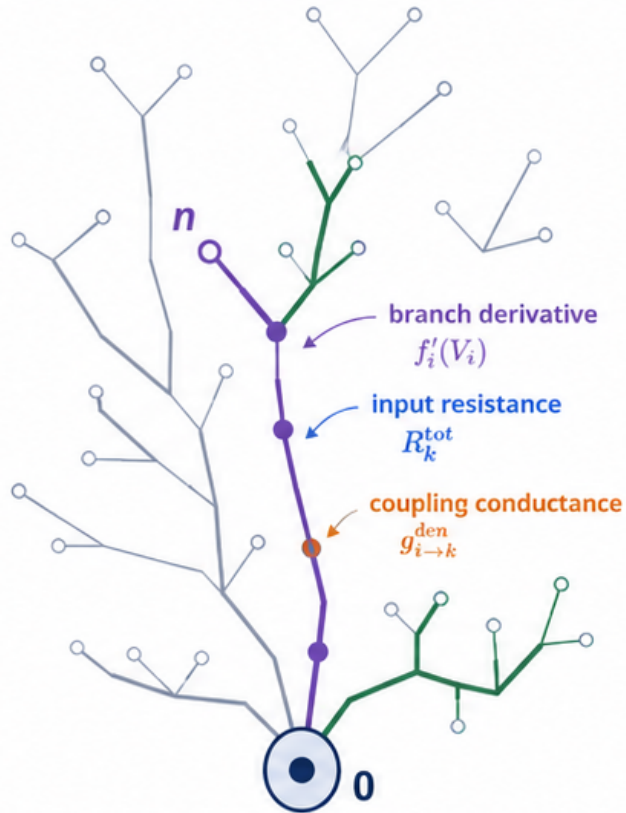
$$\frac{\partial L}{\partial g_j^I} = x_j^I R_n^{\text{tot}} (E_I - V_n) \delta_{n,u}^V, \quad j \in \mathcal{S}_n^I$$

Interpretation

- E and I have opposite local voltage sensitivities when $E_I < V_n < E_E$.
- The sign of the final loss gradient also depends on the compartment learning signal $\delta_{n,u}^V$.
- Near E_I , an inhibitory synapse can weakly drive its own current yet still change other gradients through conductance loading.

From tree transport to the general adjoint

Directed tree: single path transport of returned signal



Recursive transport along the unique path from n to 0

$$\delta_{0,u}^V = f'_0(V_0) \delta_u$$

$$\delta_{n,u}^V = \delta_{p(n),u}^V f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$$

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n$$

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}$$

$f'_i(V_i)$: branch derivative

R_k^{tot} : input resistance

$g_{i \rightarrow k}^{\text{den}}$: coupling conductance

$p(n)$ parent of node n on the path to the root
 $\text{path}(n \rightarrow 0)$ ordered set of directed edges from n to root 0
 $(n = i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_m = 0)$

General compartmental system: steady-state adjoint

$$F(V, g, x) = 0$$

steady-state residual
(nonlinear in general)

$$J_V = \frac{\partial F}{\partial V}$$

Jacobian w.r.t. V

$$J_V^T q = \nabla_V L$$

adjoint linear system

$$\frac{\partial L}{\partial g_i} = -q^T \frac{\partial F}{\partial g_i}$$

gradient of scalar objective L
(adjoint sensitivity formula)

For the standard residual convention

$$F = GV - b$$

nodal conductance matrix G , source b

$$\frac{\partial L}{\partial g_i} = x_i (E_i^{\text{rev}} - V_n) q_n$$

local sensitivity (synapse i on compartment n)

Relationship to directed-tree transport

$$q_n = R_n^{\text{tot}} \delta_{n,u}^V$$

adjoint equals input resistance
times returned signal

In a directed tree, error transport is a single path product; in a general compartmental system, it is carried by the full adjoint / Green's function.

Restricted routed local learning rule

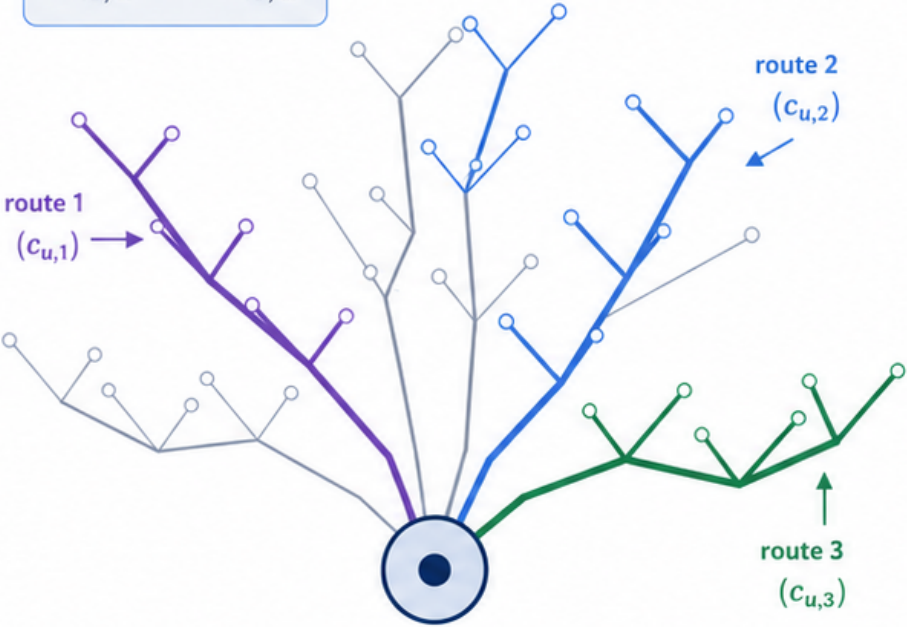
Exact and approximate local rules

Local approximate-gradient form:

$$\hat{\nabla}_{g_i} L = x_i R_n^{tot} (E_i^{rev} - V_n) \hat{\delta}_{n,u}^V$$

Exactness condition:

$$\hat{\delta}_{n,u}^V = \hat{\delta}_{n,u}^V$$



Routed teaching field (approximate local feedback field)

$$\hat{\delta}_{n,u}^V = \sum_{k=1}^K A_{u,nk} c_{u,k}$$

$A_u \in \mathbb{R}^{N_u \times K}$ route/address matrix

$\mathbf{c}_u = (c_{u,1}, \dots, c_{u,K})^\top$ communicated signed values

K feedback bandwidth (route coefficients)

N_u routed nonsomatic compartments in neuron u

Address matrix A_u (compartments $\times$ routes)

	route 1	route 2	...	route K
compartment 1	1	0	...	0
compartment 2	1	0	...	0
compartment 3	0	1	...	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
compartment N_u	0	0	...	1

Route coefficients $\mathbf{c}_u$ (length K)

$$\begin{bmatrix} c_{u,1} \\ c_{u,2} \\ \vdots \\ c_{u,K} \end{bmatrix}$$

each value is assigned to the compartments of its route

Bandwidth and routing choices: from scalar to exact compartment field

1) Layer-wide scalar	2) Neuron-specific feedback	3) Addressed routed field	4) Exact compartment field
one value shared by all neurons	one value per neuron, repeated within that neuron	several values per neuron, routed to selected subtrees	one independent value per compartment
bandwidth $K = 1$ shared across layer	bandwidth $K = 1$ per neuron	bandwidth $K > 1$ per neuron (routed)	bandwidth $K = N_u$ (exact)

Interpretation

The dendritic eligibility remains exact and local; *learning quality depends on how well the available routed field captures the exact credit field.*

Credit-operator utility predicts a conditional phase

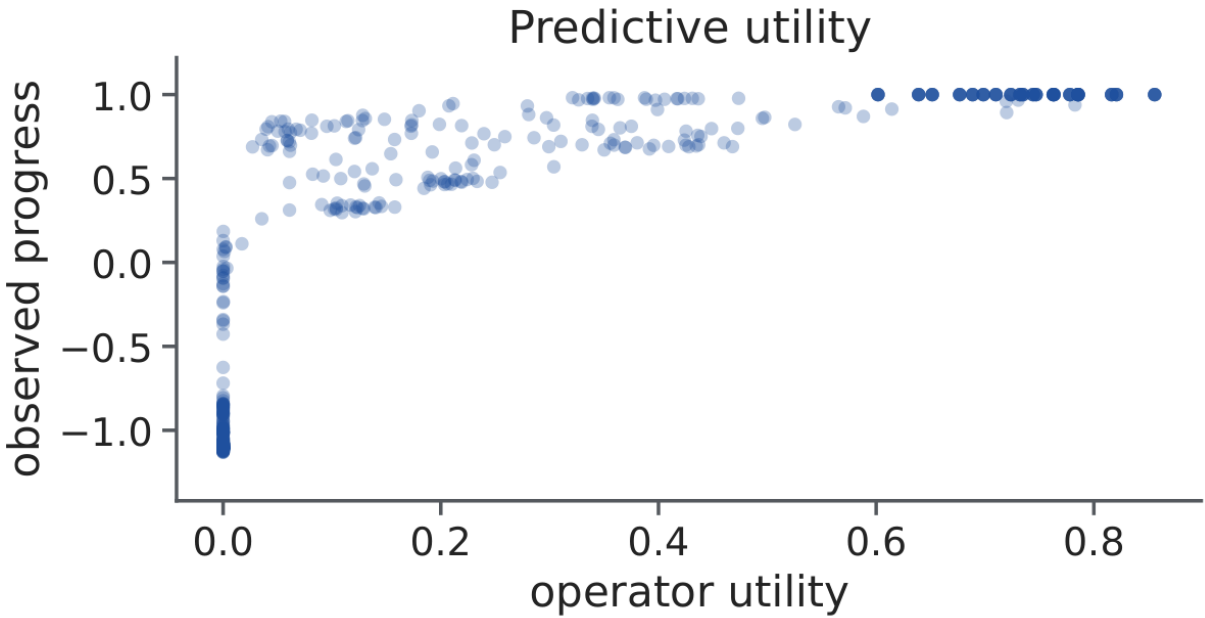
Validation uses a fixed-forward 16-coordinate Haar task: H_c is task-credit hierarchy and D_r is feedback-route resolution. The exact stochastic update has mean μ and noise ξ with covariance Σ ; M retains part of that field:

$$\tilde{\mu} = M(\mu + \xi)$$

$$U(M) = \frac{[\mu^\top M \mu]^2}{2L(\|M\mu\|^2 + \text{tr}(M\Sigma M^\top))}$$

useful signal retained
alignment of the routed update with the mean gradient

variance admitted
transported update energy and stochastic noise



$\rho_s = 0.937$ **one-step** $\rho_s = 0.916$ **final** **540 controlled conditions**

Exact for the isotropic quadratic; otherwise a smoothness bound that most directly predicts one-step progress.

alignment

route span must capture useful credit

bandwidth

benefit can peak before full rank

signal-noise

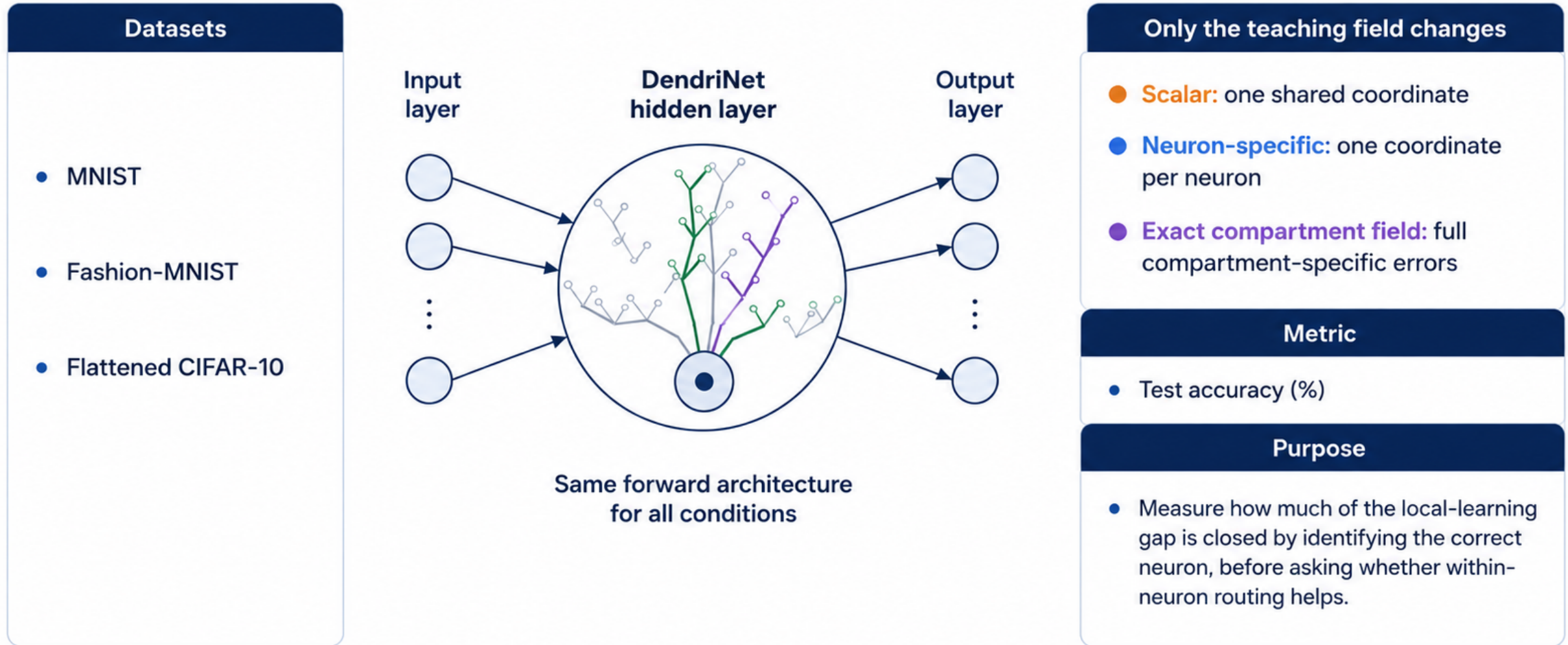
rejected noise must offset discarded signal

reliability

gain helps when it follows signal-to-noise

Operator utility balances aligned retained signal against routed update energy and admitted noise.

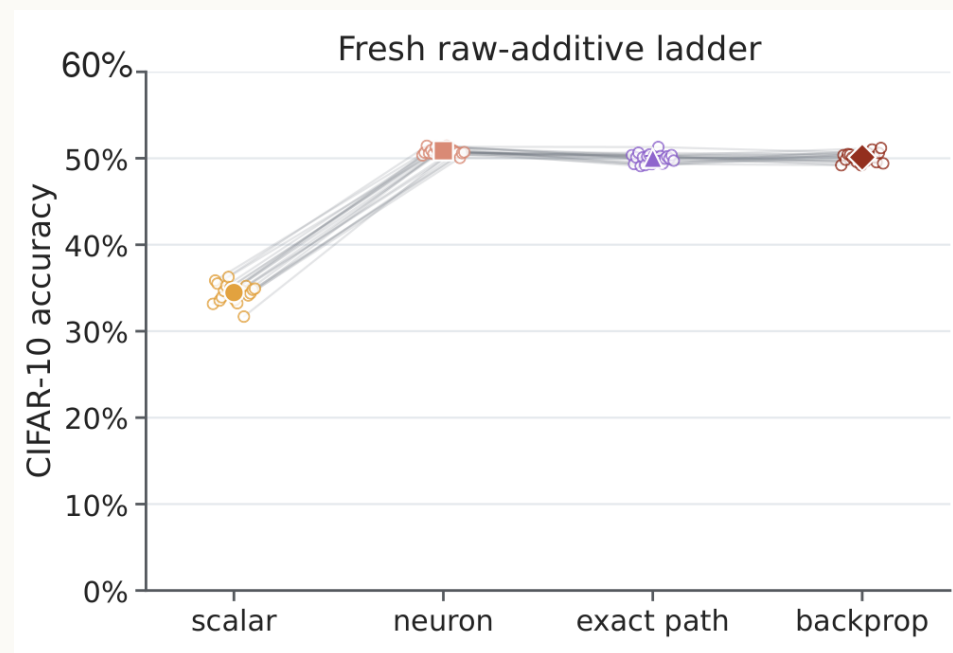
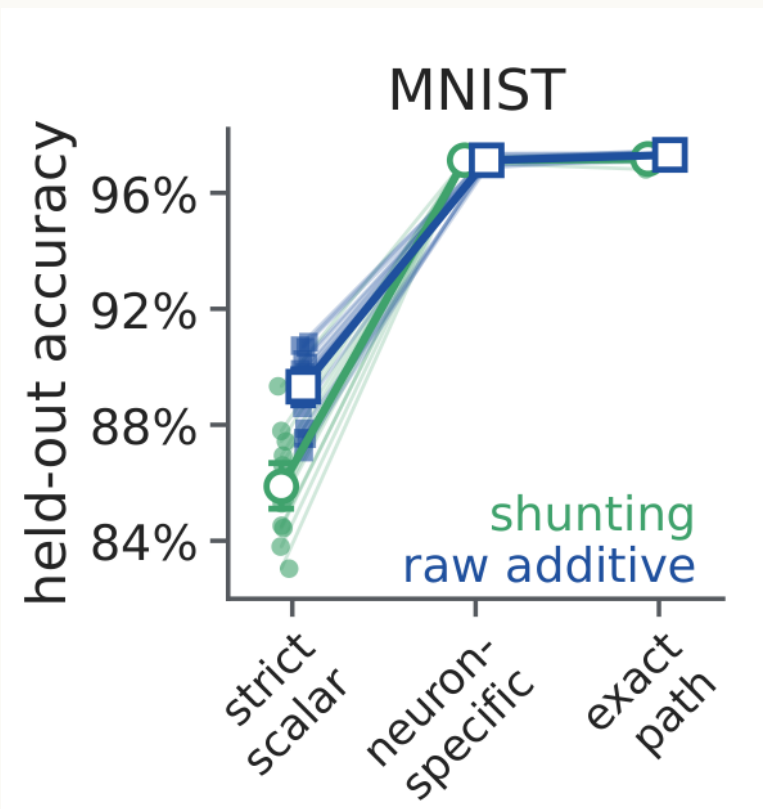
Standard benchmarks vary feedback resolution while holding the forward model fixed



Forward model, optimizer, and data are matched across conditions.

The ladder asks first whether feedback must identify the neuron, then whether it must resolve locations within its arbor.

Neuron identity closes almost the entire standard-task gap



MNIST

neuron-specific over strict scalar:

+11.2 pp shunting

+7.8 pp raw additive

exact path over neuron-specific:

< 0.2 pp

Flattened CIFAR-10

scalar 34.47%

neuron-specific 50.87%

exact path 50.01%

matched BP 50.12%

DISTAL PATH-SPECIFIC ERROR ENERGY

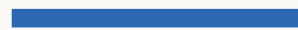
shunting



≈ 53%

Path-specific gradients exist, especially under shunting, but these benchmarks do not reward resolving them.

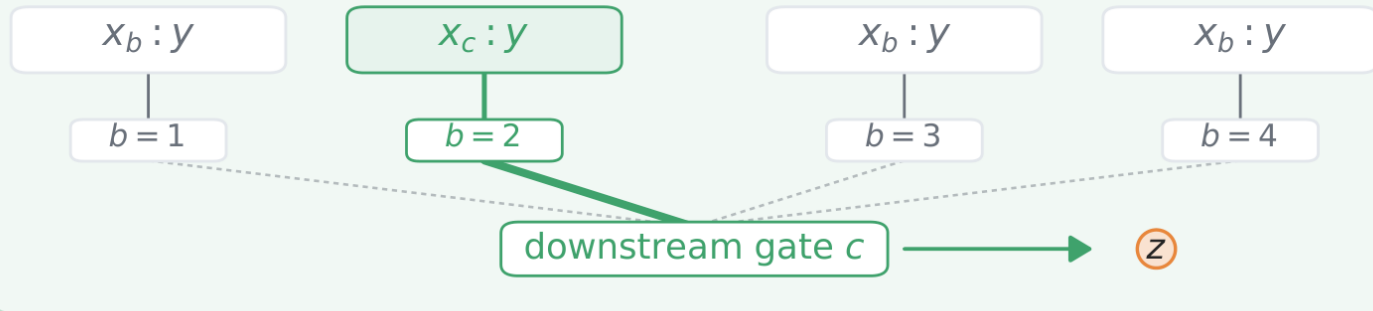
raw additive



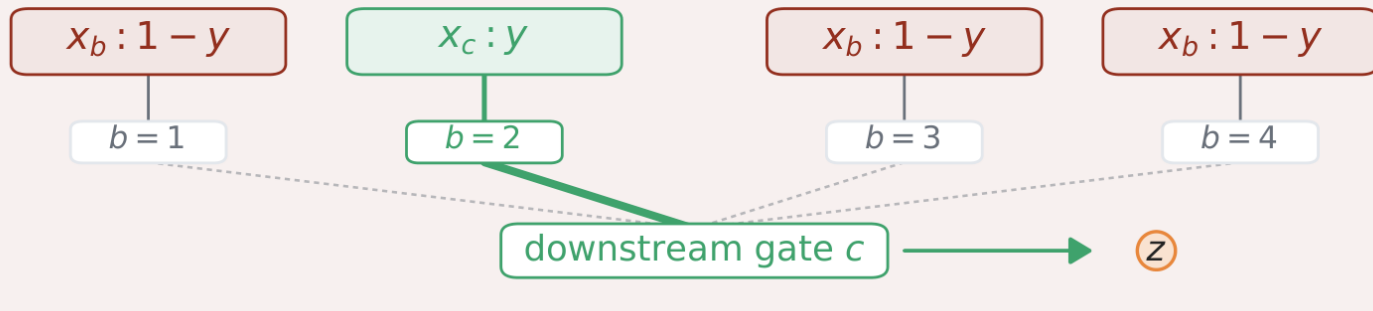
≈ 32%

Branch conflict creates a controlled need for an internal address

compatible, $\chi = 0$



conflicting, $\chi = 1$



All B branches are active. Context c selects the branch whose image class determines the label. Each nonselected branch carries the opposite-class view with probability χ .

$$\lambda_{\text{shared}} = B - 2\chi(B - 1)$$
$$\chi_c = \frac{B}{2(B - 1)}$$

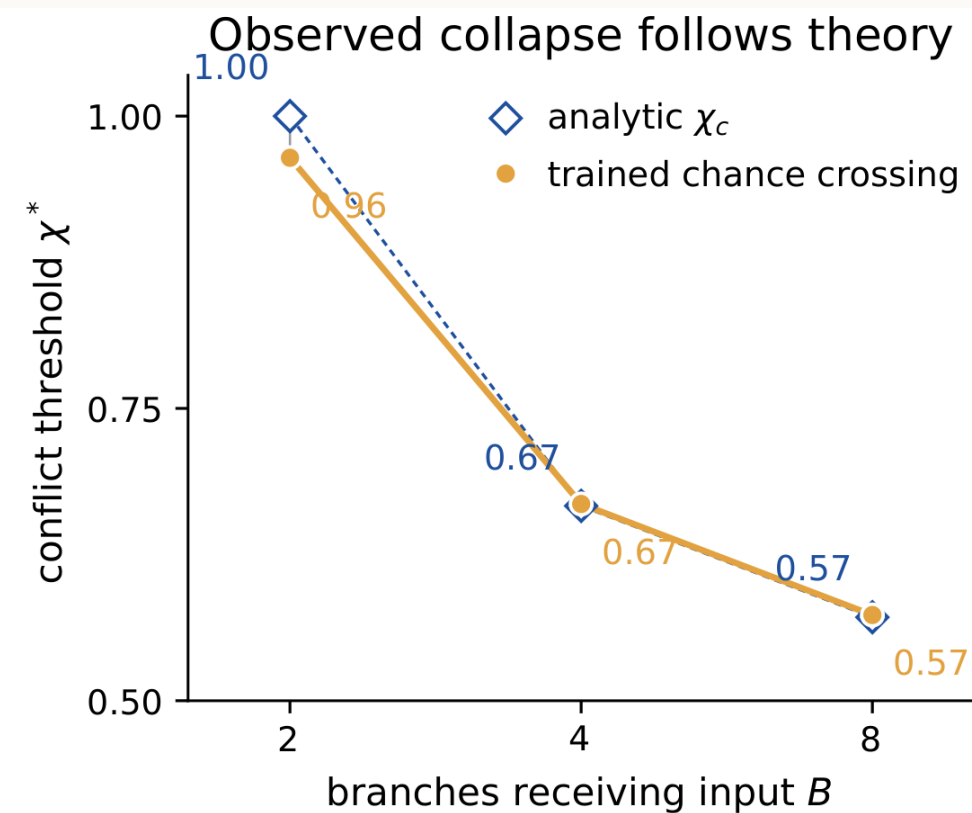
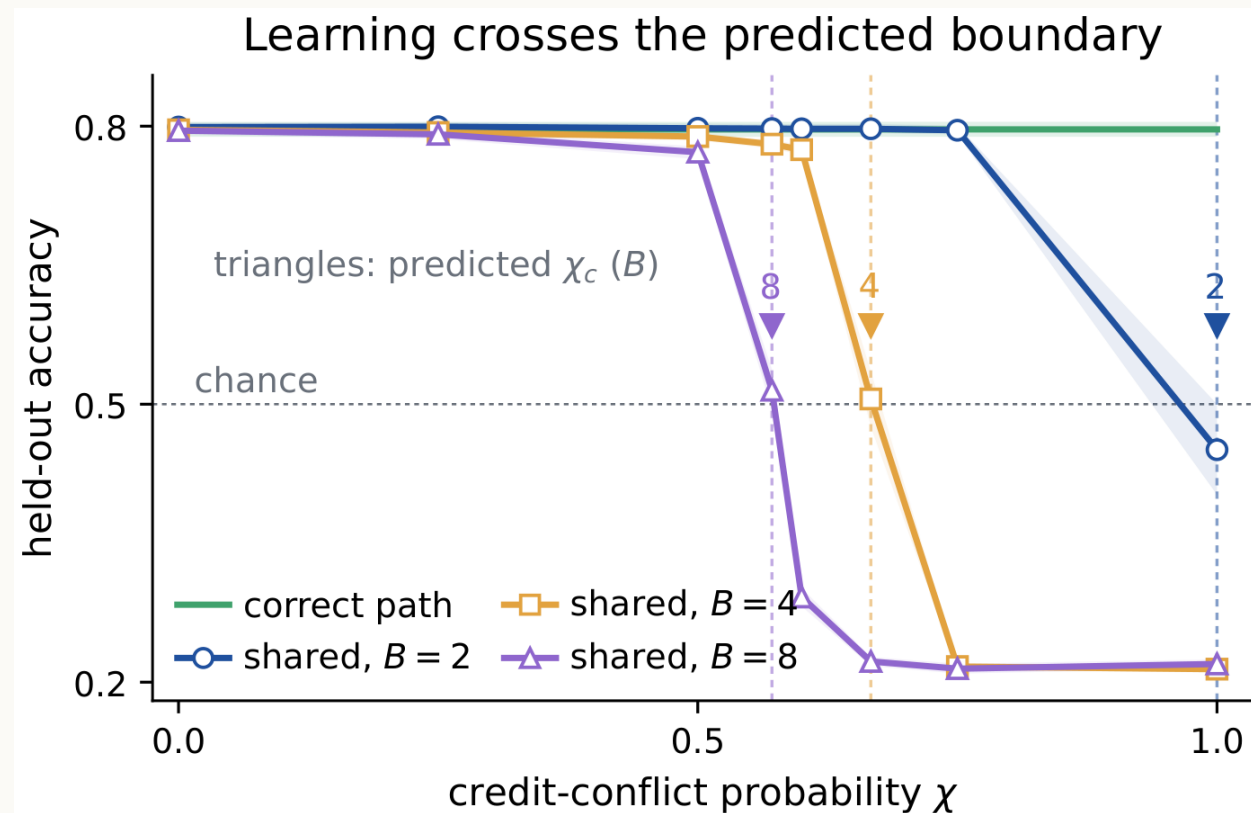
— branch-specific credit preserves the selected local sign

— a neuron-shared signal cancels above the predicted χ_c

Mechanism-matched existence test; it is not intended to represent every natural task.

The task varies one interpretable quantity: how strongly sibling branches demand opposing updates.

Shared credit fails at the predicted conflict boundary



$$\chi_c = \frac{B}{2(B-1)}$$

analytic and trained crossings agree for $B \in \{2, 4, 8\}$

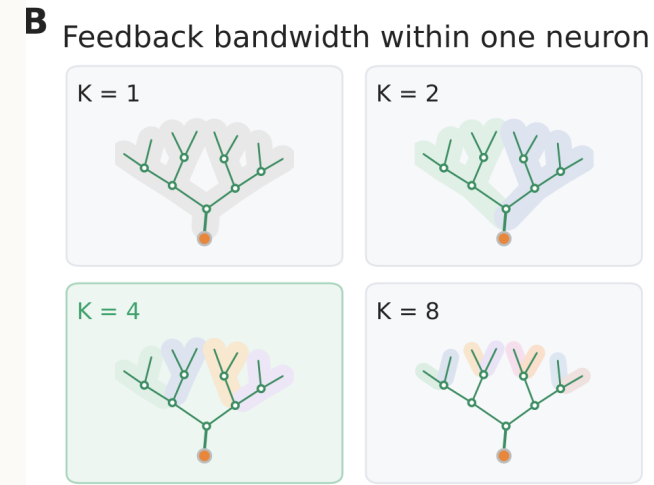
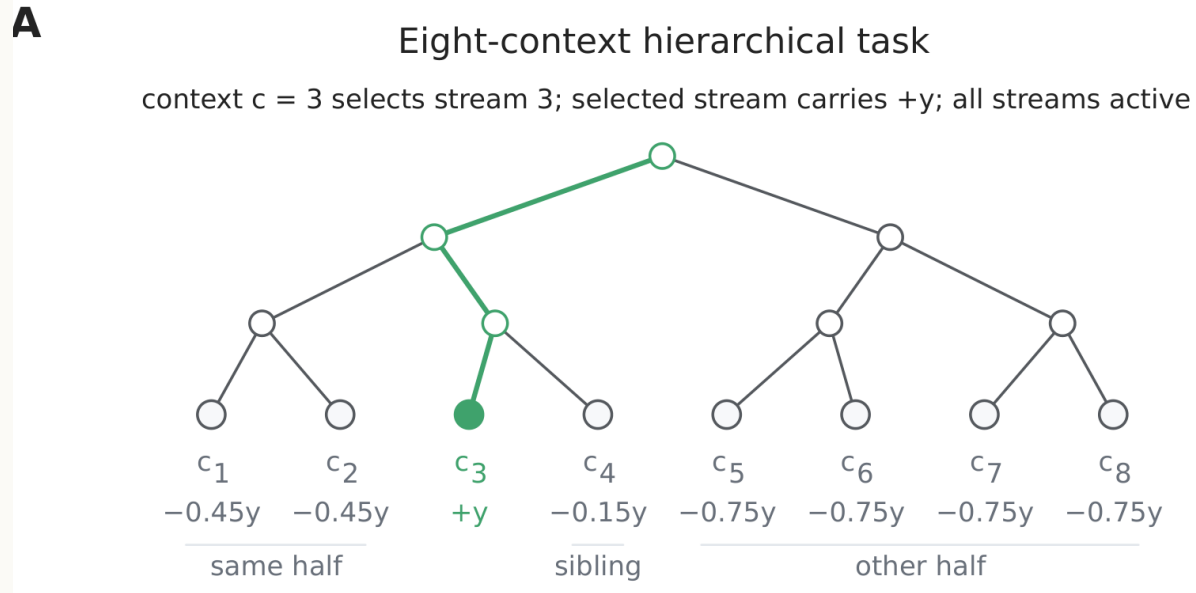
35–58 pp

correct branch route over neuron-
shared credit at full conflict

A cyclic derangement preserves rank and sparsity but sends each value to the wrong branch. It fails across 20 seeds per B .

Feedback rank is not enough: the signal must reach the branch that owns the relevant eligibility.

A hierarchical task tests subtree routes at limited feedback bandwidth



Context selects one of eight active streams independently of the balanced label. The selected stream carries the label sign; opposite-sign distractor strength grows with tree distance.

$K = 1, 2, 4, 8$ is the number of independent route coefficients within one neuron.

matched subtrees

correct supports on the known hierarchy

cyclic derangement

same routes, assigned to the wrong signals

dense low-rank basis

same feedback bandwidth, non-anatomical supports

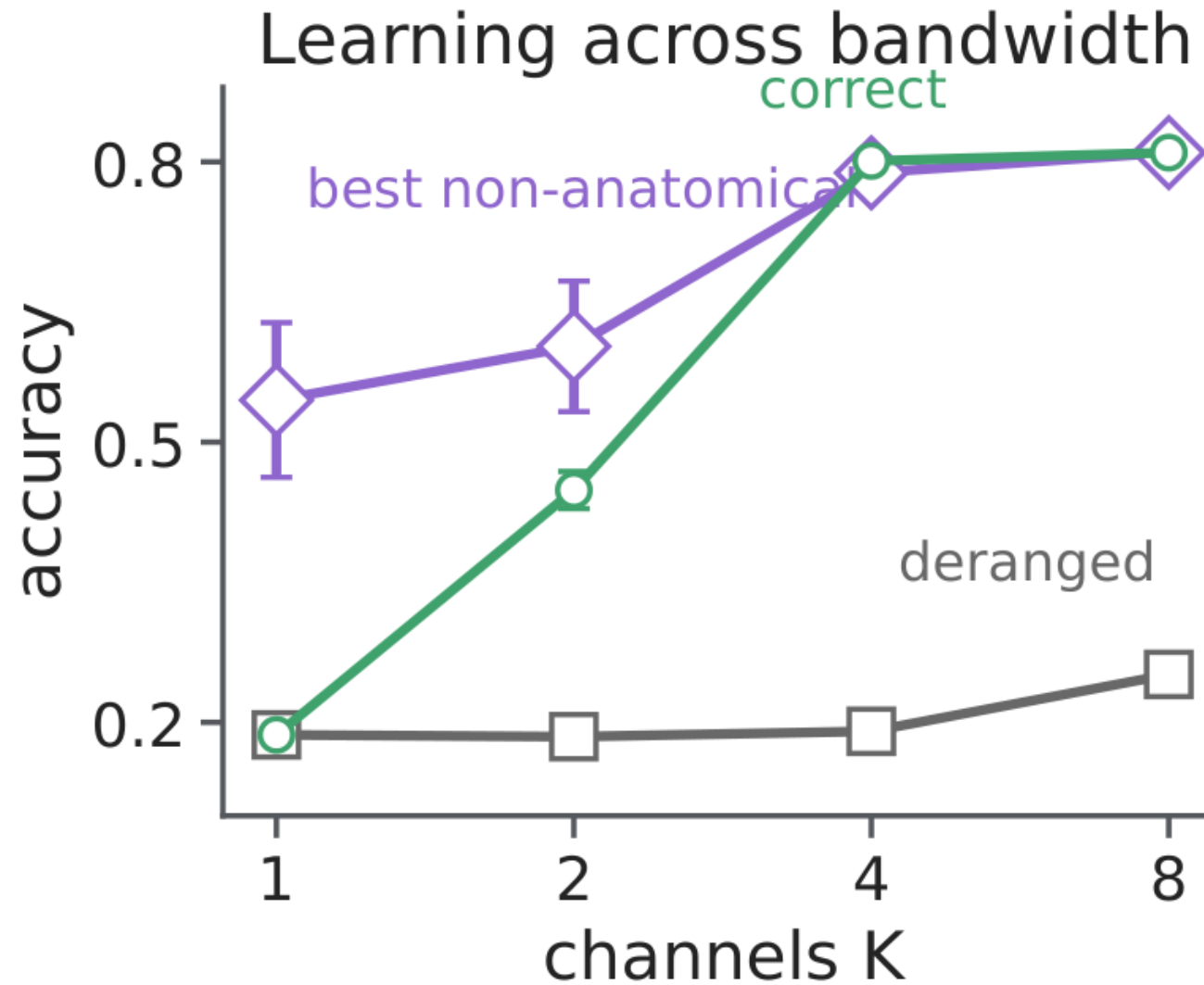
rewired tree

degree and depth preserved, ancestry changed

K is backward feedback bandwidth; it is not the number of serial dendritic stages D_p tested later.

Correct assignment is essential; ancestry adds only a narrow $K = 4$ gain

E



+61.1 pp

correct address versus cyclic derangement at $K = 4$

+1.27 pp

ancestry routes versus the strongest matched non-anatomical low-rank basis at $K = 4$

Most of the effect is correct delivery. The topology-specific increment disappears at low and full bandwidth.

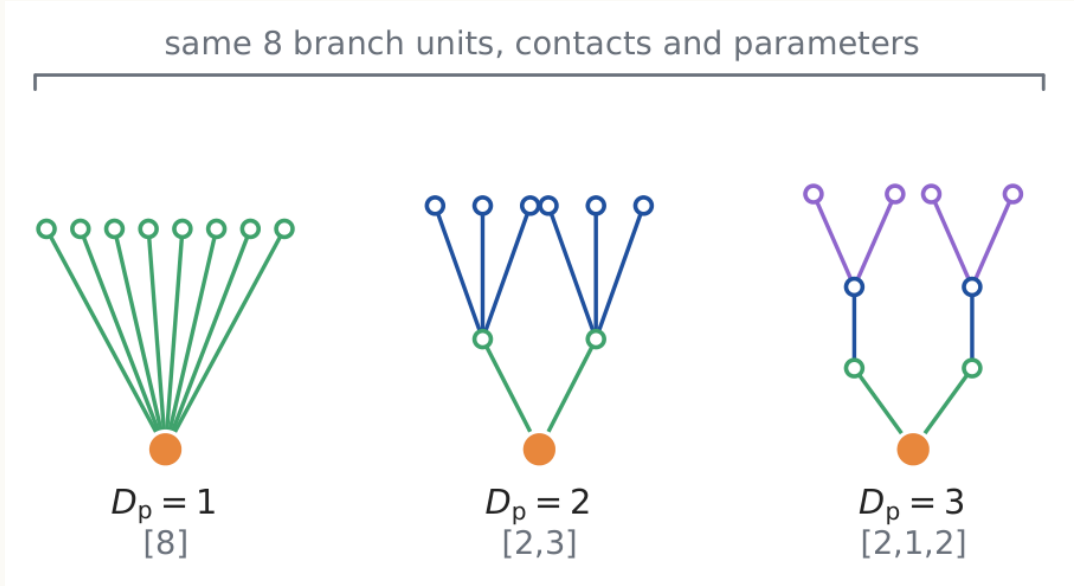
Addresses can be essential, while the extra benefit of this particular nested topology is conditional and modest.

Physical depth asks whether serial shunting matches a compositional task

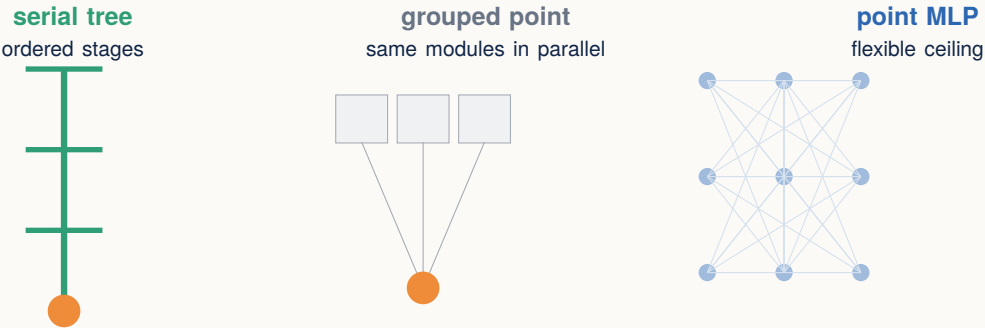
$$x_j^E = s_y \prod_{\ell=1}^{H_p} h_{\ell, a_{\ell}(j)}$$

H_p counts latent nuisance-gain levels in the task; D_p counts ordered nonlinear stages in the model.

- Nested factors** ordered cancellation of $h_1 h_2 h_3$
- Flat factors** multiplicative factors distributed across non-nested groups
- Local ratios** nuisance cancellation is available locally

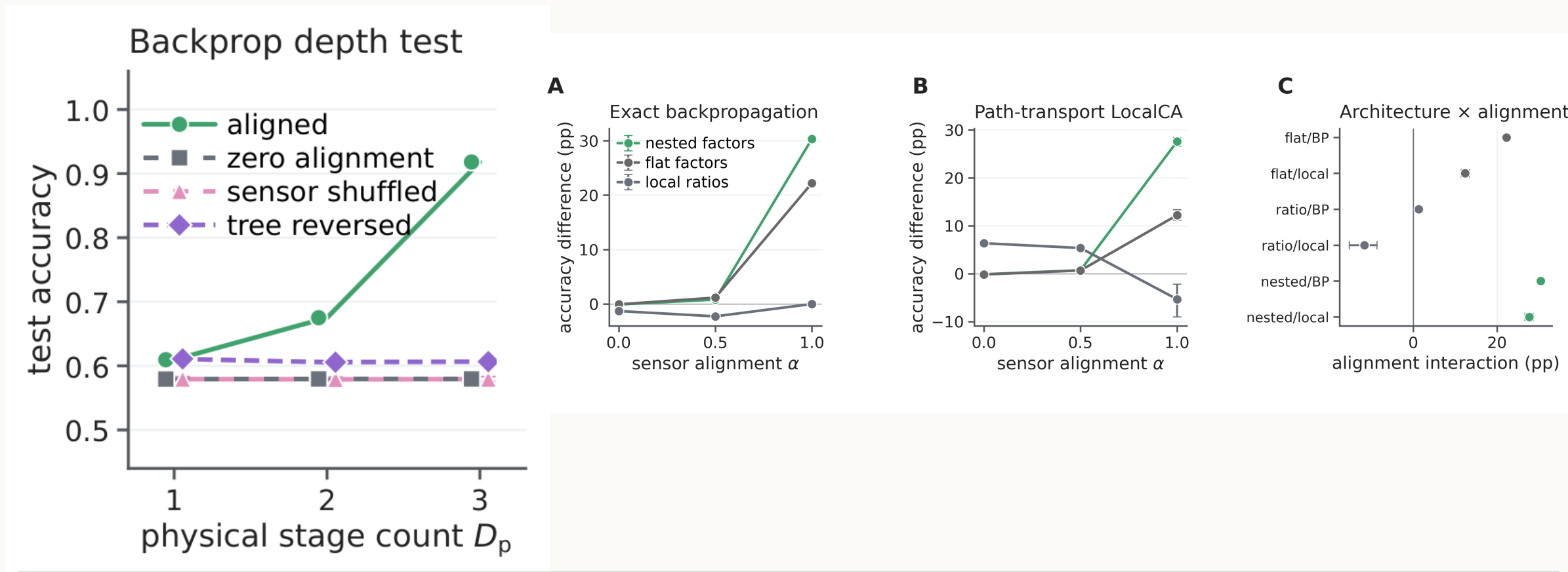


$\alpha = 0$: independent sensors; $\alpha = 1$: exact latent-gain sensors. Sensors are forward inhibitory inputs, not learning signals.



Mechanism-matched existence test at a calibrated divisive operating point; identical contacts and trainable resources for serial and grouped-point comparisons.

Serial depth helps only when operation, order, and task are aligned



+30.9 pp

aligned D3 versus D1 under BP

~11 pp

exact-path over shared-soma local credit at D3

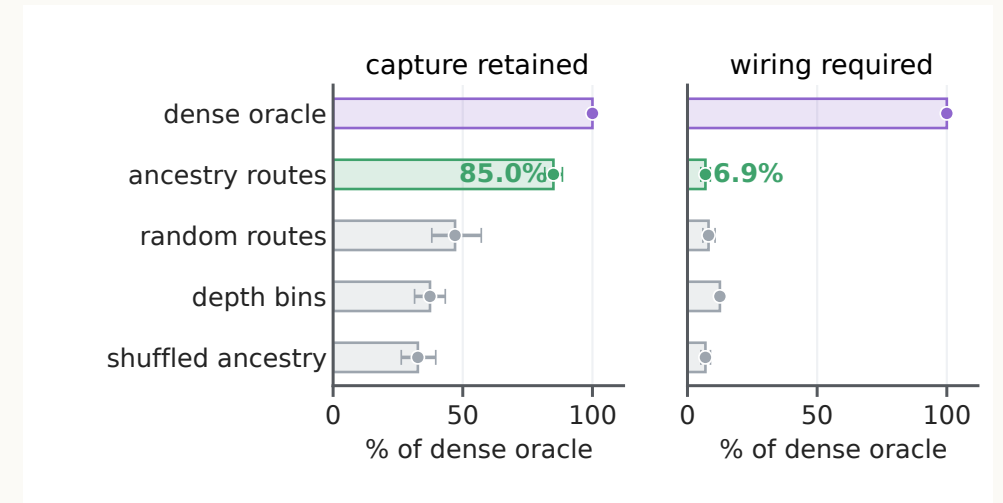
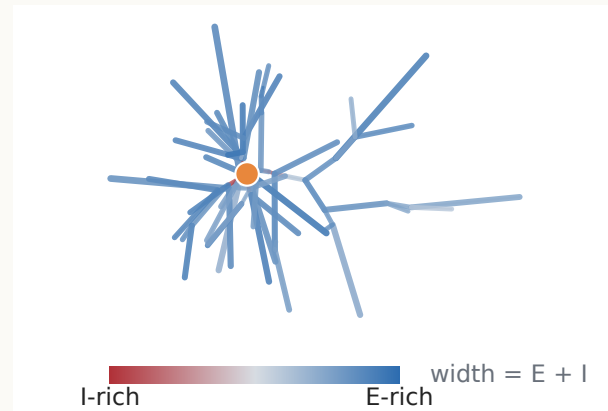
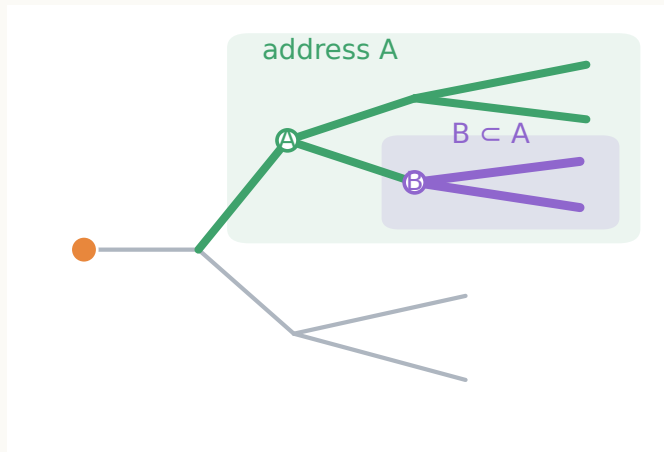
controls

reversed sensors and raw-additive depth remove the benefit

boundary

useful depth saturates; a flexible point network remains the performance ceiling

Reconstructed arbors provide sparse, predominantly coarse route dictionaries



85% capture

with about 6.9% of dense feedback connections

14.2×

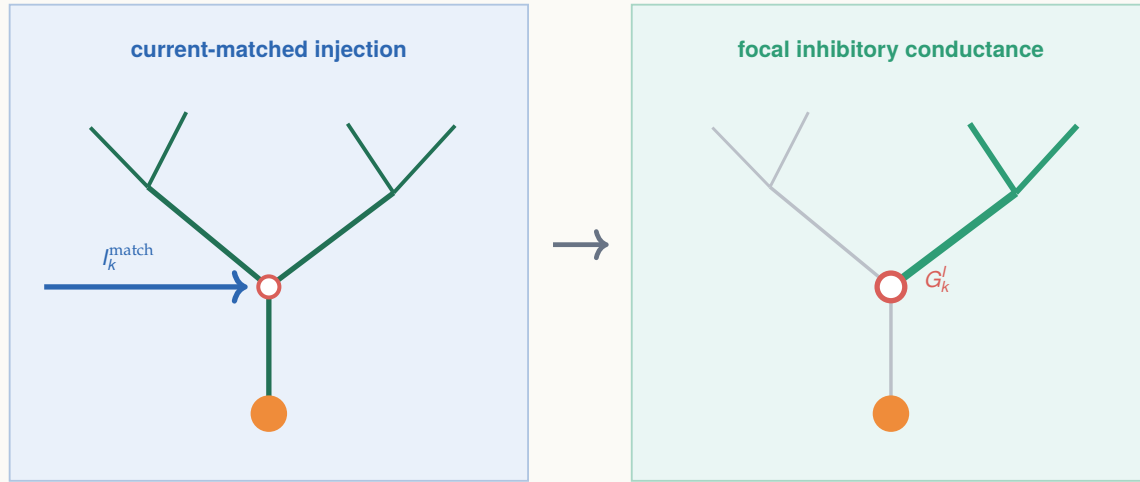
model-matched capture-per-connection ceiling

~ 2.7×

density-matched anatomy-specific advantage

Primary extension: a disjoint 47-cell cohort reproduces the route ordering. All ten qualifying cells from a second mouse show the model-matched subtree advantage. These are modeled fields on measured anatomy, not observed task gradients.

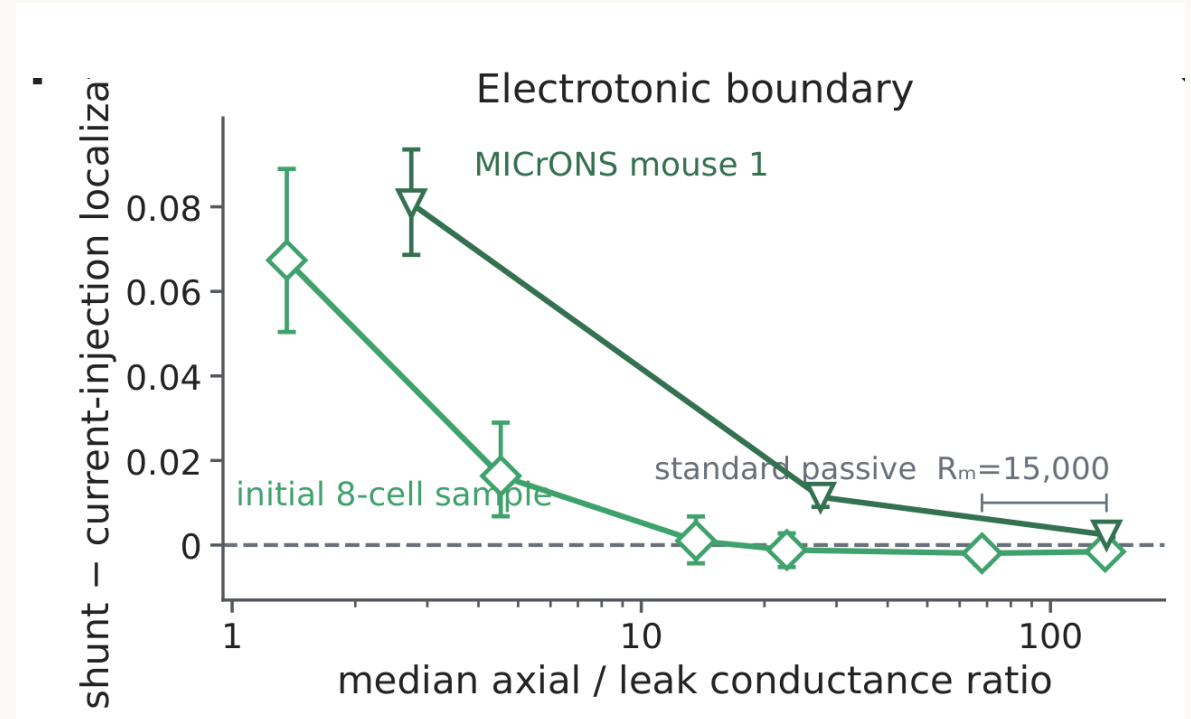
Focal shunting regulates route gain only in a permissive conductance regime



$$\frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = -R_k^{\text{tot}} \mathbf{1}[k \in \mathcal{A}(n)]$$

first-order change is restricted to descendants whose route passes through site k

Somatic voltage is restored after both interventions. Only the shunt changes G , and therefore $q = G^{-\top} \nabla_V \mathcal{L}$.



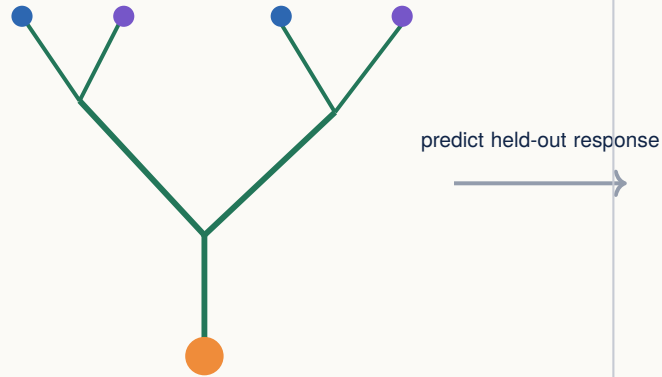
Standard passive calibration ($R_m = 15,000 \Omega \text{ cm}^2$): null.

Lower resistance / background conductance: descendant localization emerges.

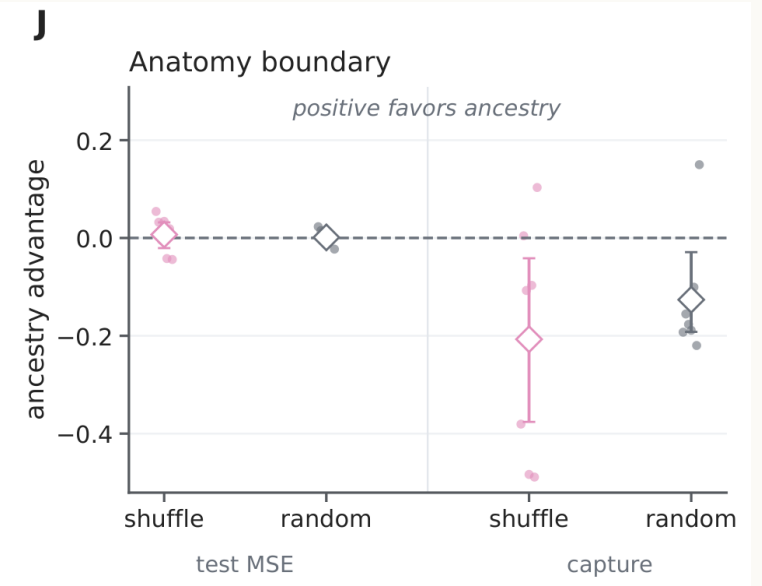
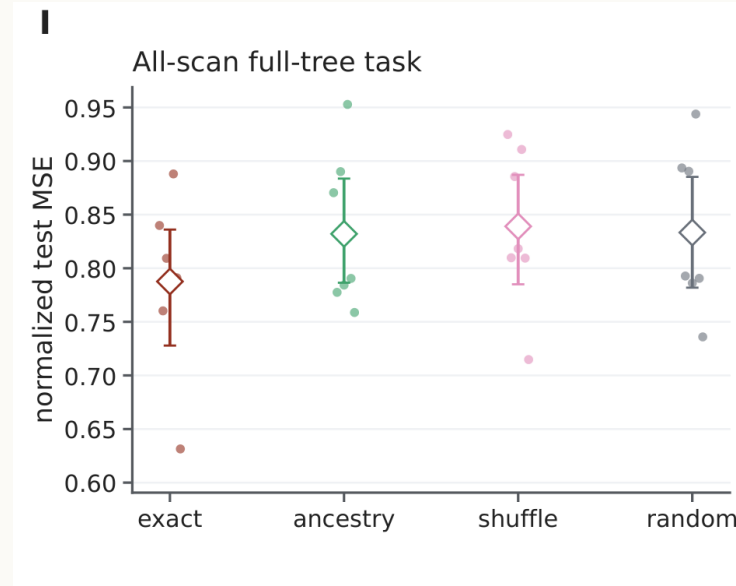
Shunting can regulate an addressed route, but spatial selectivity is state dependent rather than automatic.

Measured visual responses do not preferentially align with ancestry routes

Measured presynaptic visual responses are mapped to dendritic compartments on each reconstructed target cell. Each route dictionary predicts the target cell's held-out visual response.



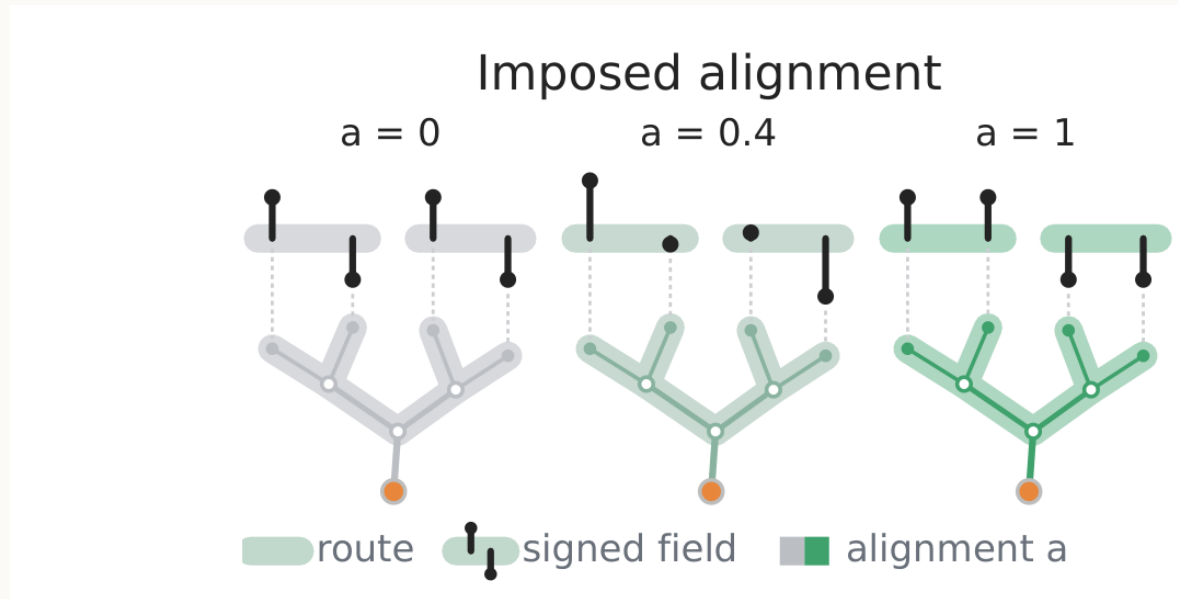
Seven target cells from one MICrONS mouse



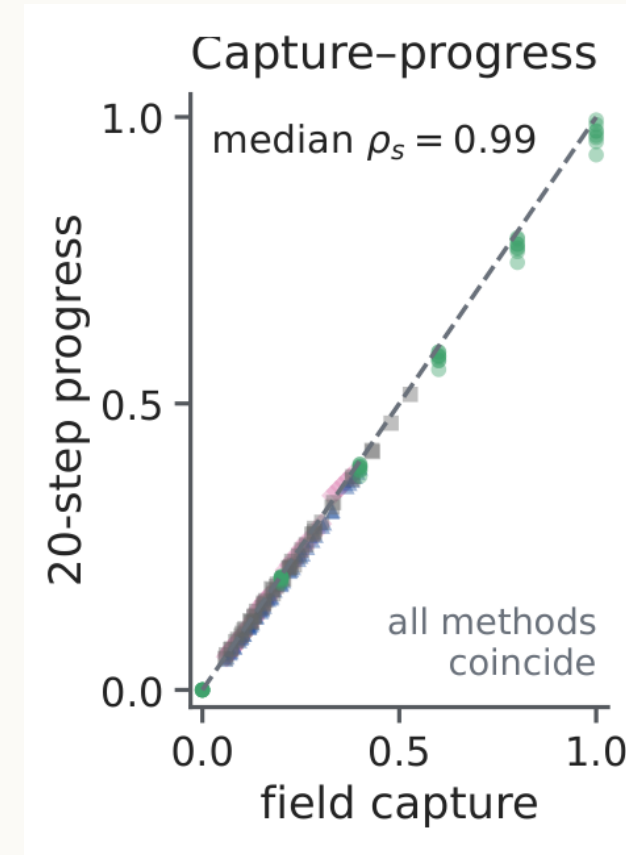
Topology-minus-matched-control effects cross zero. Morphology supplies candidate routes, but this measured task does not establish their endogenous use.

The same anatomy becomes useful when credit is rotated into its route span

$$\phi(a) = \sqrt{a} u_{\parallel} + \sqrt{1-a} u_{\perp}, \quad C_A[\phi(a)] = a$$



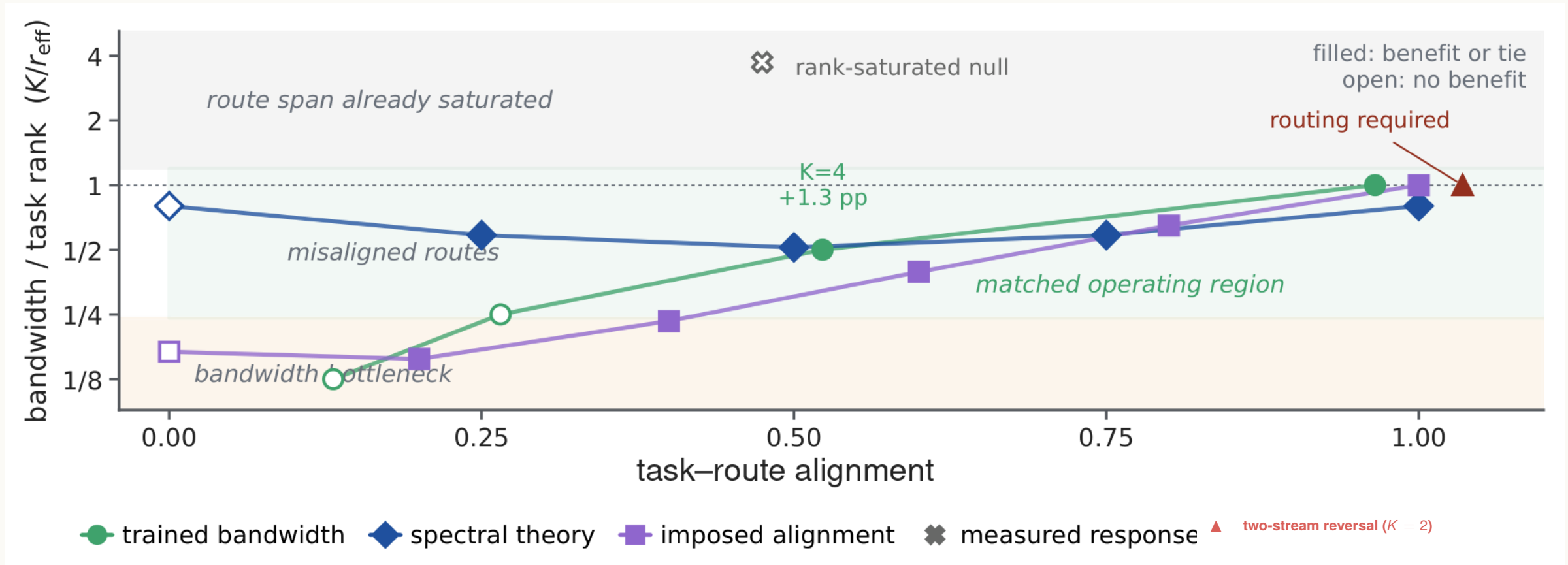
Anatomy, route energy, and curvature remain fixed; only the task-credit field rotates from orthogonal to aligned.



Alignment is imposed, not learned or observed endogenously.

This is a sufficiency test: fixed routes help when the required credit field lies in their span.

Dendritic benefit requires a match among coordinate, address, gain, and task



theoretical regime map; boundaries are not fitted

Coordinate

neuron identity closes most of the standard-task gap

Address

routes help when local credits conflict

Gain and computation

shunting and serial depth help only in matched regimes

Biological boundary

sparse capacity exists; endogenous alignment remains open

Dendrites are a conditional substrate for routing and transforming local credit—not a general replacement for backpropagation.